Labsheet 3 - Computer vision

Name – Yash kumar Srivastava
3rd year section B
roll no. 70

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
# 1. Load image
img = cv2.imread("image.jpg")
# 2. Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
# 3. Gaussian Filter
gaussian = cv2.GaussianBlur(gray, (5, 5), 0)
# 4. Median Filter
median = cv2.medianBlur(gray, 5)
# 5. Average/Mean Filter
average = cv2.blur(gray, (5, 5))
# 6. Laplacian Filter
laplacian = cv2.Laplacian(gray, cv2.CV_64F)
aplacian = np.uint8(np.absolute(laplacian))
# 7. Sobel Filter
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_x = np.uint8(np.absolute(sobel_x))
sobel_y = np.uint8(np.absolute(sobel_y))
# 8. Display results
titles = [
    "Original", "Gaussian", "Median",
    "Average", "Laplacian", "Sobel X", "Sobel Y"
]
images = [
    gray, gaussian, median,
    average, laplacian, sobel_x, sobel_y
]

plt.figure(figsize=(12, 8))

for i in range(len(images)):
    plt.subplot(2, 4, i + 1)
    plt.imshow(images[i], cmap="gray")
```

```
    plt.title(titles[i])
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Observations

Filter	Observation
Gaussian	Smooths image and reduces noise while retaining major structures.
Median	Effectively removes salt-and-pepper noise.
Average	Produces smooth image but may blur edges.
Laplacian	Enhances edges and fine details.
Sobel X	Detects vertical edges.
Sobel Y	Detects horizontal edges.

Result

The spatial filtering techniques were successfully implemented using Python and OpenCV. **Gaussian, Median, and Average filters** performed smoothing and noise reduction, while **Laplacian and Sobel filters** enhanced and detected image edges.

Questions and Answers

1. What is spatial filtering? How is it used in digital image processing?

Spatial filtering is a technique that modifies a pixel using its neighboring pixels. It uses a **kernel or filter mask** to process an image.

It is used for:

- Noise removal
- Image smoothing
- Sharpening
- Edge detection

---

2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.

Low-Pass Filters:

- Smooth the image.
- Remove noise and fine details.
- Reduce rapid intensity changes.
- Examples: **Gaussian, Average, Median**

High-Pass Filters:

- Enhance edges and fine details.
- Detect rapid intensity changes.
- Examples: **Sobel, Laplacian**

---

3. Compare Average Filter, Gaussian Filter, and Median Filter.

Filter	Working Principle	Main Application
Average	Takes the average of neighboring pixels	Basic smoothing
Gaussian	Uses weighted average based on Gaussian distribution	Noise reduction
Median	Takes the median value of neighboring pixels	Salt-and-pepper noise removal

---

4. Why is the Median Filter particularly effective for removing salt-and-pepper noise?

Salt-and-pepper noise contains very bright or very dark unwanted pixels. The median filter replaces these pixels with the **median value of their neighbors**, effectively removing the noise while preserving edges.

---

5. Explain the role of convolution kernels in spatial filtering.

A convolution kernel is a small matrix containing numerical values. It moves over the image and calculates a new value for each pixel using its neighboring pixels.

Different kernels are used for:

- Blurring
- Sharpening
- Edge detection
- Noise reduction

---

6. What is the purpose of the Sobel and Laplacian operators in edge detection?

The **Sobel operator** detects intensity changes in horizontal and vertical directions.

- Sobel X → detects vertical edges
- Sobel Y → detects horizontal edges

The **Laplacian operator** detects rapid intensity changes in multiple directions and is mainly used for edge enhancement.

---

7. Why are filtering operations considered an essential preprocessing step in computer vision?

Filtering improves image quality before further processing. It removes unwanted noise and enhances important features.

It helps improve tasks such as:

- Object detection
- Edge detection
- Image segmentation
- Feature extraction

---

8. Discuss the trade-off between image smoothing and edge preservation during filtering.

Smoothing reduces noise but excessive smoothing can blur important edges and details. On the other hand, less smoothing preserves edges but may leave noise.

Therefore, the **proper filter and kernel size** should be selected according to the application.

---

9. Mention four real-world applications where spatial filtering techniques are widely used.

1. **Medical Imaging** – noise removal and edge enhancement.

2. **Surveillance** – object and motion detection.
3. **Remote Sensing** – satellite image enhancement.
4. **Autonomous Vehicles** – road and object detection.

---

10. Compare spatial domain filtering with frequency domain filtering.

Spatial Domain Filtering	Frequency Domain Filtering
Works directly on image pixels.	Works on image frequency components.
Uses kernels and convolution.	Uses transforms such as Fourier Transform.
Relatively simple for basic filtering.	Useful for advanced frequency analysis.
Examples: Gaussian, Median, Sobel	Examples: Frequency-domain Low-Pass and High-Pass filters